

MAE 6760: Final Report

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Abstract—For modern autonomous driving of high speed vehicles, high-accuracy localization is an essential requirement for proper control and planning, especially in competition environments. More specifically, for the Shell Eco Marathon, a car that is going 20+ MPH and on an angled track needs to have high confidence in both angle and position in order to control the car properly on a banked track. This work aims to do exactly this - provide a rigid body based EKF that can provide the orientation and position data to a high level of accuracy, while maintaining model fidelity and speed at runtime. Using JAX to formulate automatically differentiable dynamics, the EKF runs at 5 ms per step, and results in an average RMSE of approximately 2.8 cm and 3.5 cm/s for position and velocity respectively.